

Architectural Particular Specification
TILING
SECTION 8.4

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Main Contract for
Proposed Data Centre Development
at 2-10 Tai Yuen Street, Kwai Chung, N.T.
Andrew Lee King Fun & Associates Architects Limited

SECTION 8.4

TILING

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. Section Includes: The Work of this Section includes, but is not limited to, the furnishing and installing of the following:
1. Floor and wall tiles, vitrified, glazed, porcelain mosaic floor tiles, quarry tiles and homogenous tiles.
 2. Setting materials and grout.
 3. Expansion joint materials.
 4. Tile cap, cove, nosing and other trim pieces.
 5. Decorative metal strips.
 6. Tile adhesive, grout and sealants.

1.02 REFERENCES

- A. British Standards
1. BS473 : 500:1990 – Specification of Concrete Roofing Tiles and Fittings
 2. BS 5385 : Part 1 : 1990 - Code of Practice for the Design and Installation of Internal Ceramic Wall Tiling and Mosaics in Normal Conditions.
 3. BS 5385 : Part 2 : 1991 - Code of Practice for the Design and Installation of External Ceramic Wall Tiling and Mosaics (including Terra Cotta and Faience Tiles.
 4. BS 5385 : Part 3 : 1989 - Code of Practice for the Design and Installation of Ceramic Floor Tiles and Mosaics.
 5. BS 5385 : Part 4 : 1992 - Code of Practice for the Tiling and Mosaics in Specific Conditions.
 6. BS 8000 : Section 11.1 : 1989 - Code of Practice for Wall and Floor Tiling : Ceramic Tiles, Terrazzo Tiles and Mosaics
 7. BS 8204 : Part 1 : 1987 - Code of Practice for Concrete Bases and Screeds to Receive In-site Floorings.
 8. BS 6431 : Part 1 : 1983 - Ceramic Floor and Wall Tiles : Specification for Classification and Marking.
 9. BS 6431 : Part 2 : 1984 - Specification for Extruded Ceramic Tiles with a Low Water Absorption ($E \leq 3\%$), Group A1.
 10. BS 6431 : Part 3 : Section 3.1 : 1986 - Specification for General Products (of Ceramic Floor and Wall Tiles).
 11. BS 6431 : Part 12 : 1983 : Method for Determination of Modulus of Rupture.
 12. BS 6431 : Part 16 : 1983 : Method for Determination of Resistance to Thermal Shock.
 13. BS 6431 : Part 23 : 1986 - Specification for Ceramic Tile Sampling and Basis for Acceptance.
 14. Construction Product List A, Part 2, 1.10 by DIBt – Notes for the application of waterproofing directly under tiles for indoor and outdoor use.

B. American Standards:

1. ANSI A108.1A - 1992 : Installation of Ceramic Tile in the Wet-Set Method with Portland Cement Mortar.
2. ANSI A108.6 - 1992 : Ceramic Tile Installed with Chemical-Resistant, Water Cleanable Tile Setting and Grouting Epoxy.
3. ANSI A108.10 - 1992 : Installation of Grout in Tilework.
4. ANSI A108.11 - 1992 : Interior Installation of Cementitious Backer Units
5. ANSI A118.1 - 1992 : Chemical-Resistant, Water-Cleanable, Tile Setting and Grouting Epoxy and Water Cleanable Tile Setting Epoxy Adhesive.
6. ANSI A118.4 - 1992 : Latex-Portland Cement Mortar
7. ANSI A118.6-1992 : Ceramic Tile Grouts
8. ANSI A136.1 - 1992 : Organic Adhesives for Installation of Ceramic Tile
9. ANSI A137.1 - 1988 : Ceramic Tile
10. ANSI A10.20 - 1988 : Safety Requirements for Ceramic Tile, Terrazzo and Marble Work.
11. Tile Council of America (TCA): Handbook For Ceramic Tile Installation, latest edition.

1.03 QUALITY ASSURANCE

A. Qualifications of Manufacturer and Installer

1. Manufacturer: Company specializing in the manufacturing of high-quality tiles and pavers meeting the specified standards herein with a minimum of ten years documented experience.
2. Installer: Company specializing in high-quality installation of tiles and pavers with a minimum of two years documented experience.

1.04 PERFORMANCE REQUIREMENTS

A. Water Absorption Requirement

1. All wall, and floor tiles and roof pavers shall be tested in accordance with ASTM C373 (Test Methods for Evaluating Water Absorption, Bulk Density, Apparent Porosity and Specific Gravity of Fired Whiteware Products) and shall be classified as:
 - a. Impervious : Tiles exhibiting 0.5% or less water absorption (Group I per BS 6431 : Part 1 classification), or
 - b. Vitreous : Tiles exhibiting more than 0.5% but not more than 3.0% water absorption (Group I per BS 6431 : Part 1 classification).
2. All roof pavers shall be classified "impervious" type pavers. Thermal shock resistance per BS 6431 : Part 16.

B. Slip-Resistance Requirements

1. Tiles and pavers on floors and roofs/decks, respectively, shall provide a safe, non-slip, walking surface for both wet and dry conditions. When floor tiles and pavers are tested to ASTM C1028 (Evaluating the Static Coefficient of Friction of Ceramic Tile and other Like Surfaces by the Horizontal Pull Method), they shall have a Coefficient of Friction of 0.6 and above.
2. Coefficient of Friction shall be greater than 0.4 in wet condition when tested in accordance to Bench Friction Apparatus (Tortus) Method per BS 5385 : Part 4.

C. Breaking Strength Requirements

1. All floor and wall tiles and pavers shall be capable of withstanding the expected load and impact ratings of specified installations. The breaking strength of tiles and pavers shall be determined by ASTM C648 (Test Method for Breaking Strength of Ceramic Tile).
2. Floor tiles and pavers shall have minimum Modulus of Rupture conforming to BS 6431 : Part 12 of 35N/mm². Minimum density = 27 kg/m².
3. Roofing tiles shall be capable of withstanding the expected load and impact ratings of specified installations. The breaking strength of six plan roofing tiles accordance to BS 473, 500:1990 clause 10.2 should be not less than 490N.
- 4.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

A. Submit the following as required in Section 1.5.

B. Shop Drawings:

1. Include drawings, details and elevations where applicable. Floor Pattern Plan and Elevation Drawings shall be drawn at 1:10 scale. Detail drawings at 1:2 scale.
2. Provide shop drawings for the Architect's review and approval.
3. Show details at all floor tile penetrations, abutments, plinths, roof to parapet and wall to floor transitions, drain and outlets, internal and exterior corners, and floor expansion and control joints installation and details.
4. Submit a 600 x 600 sample board of installed tiles with patterns and grout joints as specified for the Project.

C. Samples:

1. Samples of each type and color of tiles and pavers, special cove, nosing, capping and corner tile trim units specified herein. The cost of postage and transportation and all samples shall be the responsibility of the Contractor.
2. Samples of corner cove tiles for glass accent strips.
3. The Contractor shall submit adequate samples for testing (if required) by Testing Laboratory prior to any installation.
4. Submit colour grout samples for Architect's selection.

D. Submit technical literature for proposed tile setting bed/adhesive for approval. The type of adhesive proposed to be in strict accordance with the tile manufacturers recommendations.

E. Submit Statement of Manufacturer's Review Form prior to any installation Work as required in Section 01 78 00.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE, AND HANDLING

- A. Deliver all materials in Manufacturer's original unopened containers, dry, and undamaged with labels and production batch number intact.
- B. Check to make sure that tiles and pavers are not damaged. Remove and replace all damaged products.
- C. Store materials in weather protected environment, clear of ground and moisture.
- D. Setting materials and grout shall be prevented from water and weather exposure, and they shall be new, unexpired materials. All grout materials to be from a single batch of manufactured material.

1.10 MOCK-UPS

- A. This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.
- B. The work under the Contractor's direct supervision, employing workmen as they would be employed during the actual erection at the Project Site. All mock-ups costs shall be deemed to be included in the Contract Sum.
- C. The mock-ups shall be full size and represent, as closely as possible, conditions which will exist in the completed Project Building.
- D. The Contractor shall ensure that all necessary shop drawings, advance orders and deliveries are arranged to allow completion of the sample mock-up areas.
- E. Production for final job site erection shall not start until approval has been obtained from the Architect.
- F. Acceptance of Mock-ups as final in placed Work shall be at the sole discretion of the Architect.

1.11 TESTING

- A. Before commencing any Tiling Works, the Contractor must demonstrate on Site that all adhesive and materials are fully compatible with each other, including as-installed waterproofing system. Unless otherwise specified, a mock-up panel of size not less than 2m x 2m on selected tiling works (to be selected by Architect) shall be carried out by the contractor for the inspection and approval by the Architect prior to bulk ordering of the selected tiles / finishes. No work shall be undertaken when the surface moisture exceeds the permissible, as tested by the moisture testing equipment on Site. The spreading of adhesive over large areas before placing of membrane is forbidden. Keep record of the adhesive used and check against the agreed spreading rate of membrane. Bubbles formed in the membrane must be made good in accordance with the Manufacturer's technical installation literature.

B. Test Reports:

Submit a certified laboratory test report in accordance with ZDB leaflet, Technical Munich University to verify the compatibility between all adhesive and waterproofing materials.

Submit photographic records of the test arrangement on site for the compatibility between all adhesive and materials including waterproofing system, set-up and of the performance of the mock-ups, and constituent components of the same, during fabrication, assembly and under test. Submit dated photographic records one set each to the Architect and the Waterproofing Manufacturers.

Provide copies of the agreed waterproofing shop drawings for recording any modifications found necessary during and after the laboratory tests. The testing laboratory shall accurately and neatly record all changes, revisions, modification etc., made to the shop drawings. At the completion of the testing, one copy each of the mark-up Drawings shall be given to the Contractor, who shall submit copies to the Architect, Consulting Engineers and the Waterproofing Manufacturer.

Subsequently, and before any installation Work commence on the Project Site, the Contractor shall provide the Architect the approved shop drawings incorporating the modifications made.

All retesting, and remedial or replacement work shall be paid by the Contractor and included in the Contract Sum.

C. In-Situ Pull Out Test

1. Minimum Tensile Adhesion / Bond Strength;

- i. **0.5N/mm²** or above
- ii. The tensile adhesion strength of each of the test samples shall be not less than the minimum requirement of tensile adhesion strength for tile adhesive in accordance with EN 12004 : 2001 (EN 1348 : 1997) or DIN 18515-1 : 1998, after the tiles are fixed for not less than 28 days.

2. Minimum Time Limit: In-situ pull out tests shall be carried out after the tiles are fixed and grouted for not less than 28 days, i.e. after the primer coat, wall render and the mortar bedding has sufficient set and hardened;

3. Control Sample: A control sample panel for each type of tiles shall be set up prior to actual full scale fixing of tiles / tile construction;

- i. Minimum panel size: 2m x 2m
- ii. For each type of tiles, 6 test samples shall be tested. If any of the tests fails (i.e. to meet the minimum required tensile adhesion strength, and the mode of failure is due to the tile adhesive system and / or workmanship), 3 further tests shall be carried out in the vicinity of the unsatisfactory test samples;
- iii. If any one of these further tests fails, the control sample shall be rejected. Further control sample panel shall be set up until satisfactory results are obtained;

4. Frequency: For the actual test on the tiled walls / tile construction, in-situ pull out test shall be carried out in a minimum of 3 stages during the course of tiling;
5. No. per area: For each stage and each type of tiles, a minimum of a set of 3 test samples shall be taken for every 50 sq. m of external tiled wall. or 12 test samples, whichever is the greater;
 - i. If any of these samples fails the Contractors shall be required to carry out 3 further tests in the vicinity of the unsatisfactory test samples. If the samples fail again, reinstatement of the tiled areas is necessary. Test again until all these 3 tests meet the minimum required tensile adhesion strength.
6. Cost: The Contractor shall be responsible for the cost of all the above tests including all the further tests mentioned in items (3) & (5), the reinstatement of the tiled areas after such tests and rectification of the defective tiled areas at his own cost.

D. In – Situ Hammer Tapping Test

Minimum requirement : No “hollow” sound produced by tapping any one of the tile surface with a light hammer or a metal “wand”.

Minimum time limit : In-situ hammer tapping tests shall be carried out after the tiles are fixed and grouted for not less than 28 days.

Frequency : 5 points on each tile. (4 corners and 1 centre points of each tile)

No. per area : For each type of tiles, a minimum of a set of 3 test samples (different locations from In-situ Pull Off Test) shall be taken for every 100 sq.m of tiled floor and external wall.

If any of these tiles fails, the contractors shall be required to re-instate the tile areas and to test again until all these areas meet the minimum requirement as mentioned above.

The Contractor shall be responsible for the cost of all the above tests including all the further tests mentioned in above, the reinstatement of the tiled areas after such tests and rectification of the defective tiled areas at his own cost.

E. Infra-red Thermographic Scanning

Testing Methodology : Interim Technical Guidelines on The Inspection, Assessment and Repair of Buildings for The Building Safety Inspection Scheme (BSIS) August 1998, published by Buildings Department and ASTM D4788 are used as guidelines and reference at the survey, where applicable.

Minimum Requirement : No possible void found for the tile surface.

Minimum time limit : Infra-red Thermographic Scanning shall be carried out before the end of Defect Liability Period.

Locations : All external wall tiled areas

If any void is found, the contractors shall be required to re-instate the tile areas and to test again by Hammer Tapping Test until all these areas meet the minimum requirement as mentioned above.

The Contractor shall be responsible for the cost of all the above tests including all the further tests mentioned in above, the reinstatement of the tiled areas after such tests and rectification of the defective tiled areas at his own cost.

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. The contractor shall warrant for the materials, workmanship and performance of the proposed system(s) to the architect for review.
- C. The warranty shall also warrant that any defective materials, construction, workmanship revealed during the warranty period shall be made good at the contractor's own expense to the satisfaction of the architect.
- D. No bulk materials and systems ordering work shall proceed prior to the architect's confirmation of no adverse comment on the submitted warranty content and format. It is the contractor's sole responsibility to ensure that the required materials or systems can be ordered according to the contractor's master programme. No claim for any additional time or cost due to any delayed submission of a satisfactory guarantee will be entertained.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURES/ SUPPLIERS

A. Subject to compliance with the Project Contract Documents, provide tile grouts, adhesive, sealants and coatings from one or more of the following approved manufacturers/ suppliers, or equivalent approved by the Architect:

1. Tech Link Construction Engineering Ltd.
Unit 802, 8/F, Laford Centre,
No. 838 Lai Chi Kok Road,
Cheung Sha Wan Kowloon
Tel: (852) 22747038
Fax: 22747039
Email: kenuskwong@techlinkcon.com.hk
2. or approved equivalent.

2.02 MATERIAL

A. General

1. Refer to the finishing schedule for the modular size of the tiles. Colour, pattern and finishes to be confirmed by the Architect. The contractor can submit an equivalent or equal system or product to the Architect for approval. No claim for any additional time for the use of any alternative as proposed by the contractor will be entertained by the Architect. Any cost saving as a result of the Architect's approval for any alternative system or product shall be adjusted according to the contract conditions.

B. Ceramic Tiles

1. All ceramic tiles shall be manufactured with pure fine clay with a single fire process to a temperature of around 1200°C. All tiles shall be in compliance with the following requirements or equivalent standard.
2. Unless specified separately, general properties for ceramic tiles shall comply the following standard:

Characteristic	Standard	Requirement
Length / Width	EN 98	± 0.6%
Thickness	EN 98	± 0.5mm
Warpage	EN 98	Max. 0.5mm
Rectangularity	EN 98	± 0.6%
Water Absorption	EN 99	Max. 18%
Thermal Shock	EN 104	Pass
Chemical Resistance	EN 122	Min. Class B
Staining Resistance	EN 122	Min. Class 2
Crazing Resistance	EN 105	Pass

3. Ceramic Wall Tiles (**T-02 & T-04**)

- a. Origin/Brand : “STONE PROJECT” or approved equivalent
- b. Texture: Matt, Limestone textured, Anti-slip, min. R9
- c. Color: Beige
- d. Size : 1200 x 600, 300 x 1200, 300 x 600 (Refer to drawings)
- e. Thickness:10 mm
- f. Technical features:

Characteristic	Standard	Requirement
Water Absorption	UNI EN ISO 10545/3	Complied $\leq 0.5\%$
Modulus of Rupture	UNI EN ISO 10545/4	$\geq 50\text{N/mm}^2$
Breaking Strength	UNI EN ISO 10545/4 ASTM C648	$\geq 2000\text{ N}$ 400Lbs
Frost Resistance	UNI EN ISO 10545/12 ASTM C1026	Complied Unaffected
Chemical Resistance to high and low acid and basic concentration	UNI EN ISO 10545/13 ASTM C-650	A – LA – HA Unaffected
Staining Resistance	UNI EN ISO 10545/14 CTI 81-7D	5 No evident variation
Slip Resistance	DIN 51130	R9
	DIN 51097	A

4. Ceramic Wall Tiles (T-03)

- a. Origin/Brand : "MARGRE - PRESTIGE" or approved equivalent
- b. Texture: Matt, Limestone textured, Anti-slip, min. R9
- c. Color: Beige
- d. Size : 300x600
- e. Thickness: 10 mm
- f. Technical features:

Characteristic	Standard	Requirement
Thickness	ISO 10545 - 2	± 5%
Water Absorption	ISO 10545/3	Complied ≤0.5%
Modulus of Rupture	ISO 10545/4	≥50N/mm ²
Breaking Strength	ISO 10545/4	≥1300 N
Linear Thermal Expansion Coefficient	ISO 10545-8	Manufacturer's Standard = 6.5 x 10 ⁻⁶
Frost Resistance	ISO 10545/12	Complied
Chemical Resistance to high and low acid and basic concentration	Cahier CSTB 3735	Class A/B No Alteration
Staining Resistance	ISO 10545/14	Class 4 – 5 Simple Removal
Thermal Shock Resistance	ISO 10545/9	Complied, no crack & crazing
Shade Variation	ANSI A137.1	V2 Slight Variation

5. External Ceramic Wall Tiles

- a. Origin/Brand : "GIBO" or approved equivalent
- b. Texture: Glazed
- c. Color: Dark Grey and Off White
- d. Size : 195 x 45
- e. Thickness: 8 mm
- f. Technical features:

Characteristic	Standard	Requirement
Water Absorption	JIS A 1509-3:2008	Coefficient of water absorption Average 0.15%

Modulus of Rupture	JIS A 1509-4 : 2008	Average $\geq 50\text{N/mm}^2$
Breaking Strength	JIS A 1509-4 : 2008	Average $\geq 1300\text{ N}$
Crazing Resistance	JIS A 1509-8 : 2008	No crazing
Chemical Resistance to high and low acid and basic concentration	JIS A 1509-10 : 2008	Class A No visual change
Resistance to Surface Abrasion	JIS A 1509-6 : 2008	Abrasion stage: Failure visible at revolution Class 3

6. Approved Supplier or equivalent approved by the Architect

Arnhold & Co., Ltd.
6/F Victoria Centre, 15 Watson Road, Hong Kong
Tel: (852) 2806 4521 / 90905009
Fax: (852) 2512 9986
Email: raymondchan@arnhold.com.hk

C. Homogeneous Tiles

- a. All homogeneous tiles shall be fully vitrified glazed or unglazed surface manufactured from very pure and powder or granulose material under high pressure pressed system. All internal floor tiles shall be high absorptive resistant and mechanical strength with a low porosity in compliance with the following requirements or equivalent standards.
- b. Tile contains at least 10% pre-consumer recycled material.
- c. Nosing tiles, coved skirting tiles shall be from the same brand and series and be supplied by the same supplier of the homogenous tiles.
- d. General Properties:

Characteristic	Standard	Requirement
Dimensions and Surfaces		
Length and width	ISO 10545-2	+/- 0.6 %, max 2.0mm
Thickness	ISO 10545-2	+/- 5 %, max 0.5mm
Straightness of edges	ISO 10545-2	+/- 0.5 %, max 1.5mm
Flatness	ISO 10545-2	+/- 0.5% ,max 2.0mm
Physical Properties		
Water absorption	ISO 10545-3	≤0.5%
Breaking strength	ISO 10545-4	>35N/mm ²
Linear thermal expansion coefficient	ISO 10545-8	Manufacture's standard ≤9 x 10 ⁻⁶
Thermal shock resistance	ISO 10545-9	No crack & crazing
Chemical Properties		
Resistance to acids and bases	ISO 10545-13	ULA and UHA
Colours resistance to light	DIN 51094	ULA and UHA
Stain resistance	10545-14	Min. Class 3
Slip Resistance		
Slip Resistance	DIN 51130	R10 for internal use
		R12 for external use

2. Approved Supplier or equivalent approved by the Architect

Arnhold & Co., Ltd.
 6/F Victoria Centre, 15 Watson Road, Hong Kong
 Tel: (852) 2806 4521 / 90905009
 Fax: (852) 2512 9986
 Email: raymondchan@arnhold.com.hk

D. Porcelain Tiles

1. Porcelain Floor Tiles (**T-01**)

- a. Origin/Brand : "CERAMICHE PIEMME" or approved equivalent
- b. Texture : Semi-glazed, Anti-slip, min. R9
- c. Color : Grey Marble Patterned - 'Supreme Grey' or approved equivalent
- d. Size : 600x600
- e. Thickness: 9 mm
- f. Technical features:

Characteristic	Standard	Requirement
Length and Width	ISO 10545.2	$\pm 0.6\%$ ($\pm 2\text{mm}$)
Thickness		$\pm 5\%$ ($\pm 0.5\text{mm}$)
Warpage		$\pm 0.5\%$ ($\pm 1.5\text{mm}$)
Wedging		$\pm 0.5\%$ ($\pm 2\text{mm}$)
Flatness		$\pm 0.5\%$ ($\pm 2\text{mm}$)
Water Absorption	ISO 10545.3	$\leq 0.5\%$
Bending Strength	ISO 10545.4	$>35\text{N/mm}^2$
Linear Thermal Expansion between 20°C and 100°C	ISO 10545.8	Manufacturer's Standard $= 7 \times 10^{-6}$
Thermal Shock Resistance	ISO 10545.9	Required, No crack & crazing
Frost Resistance	ISO 10545.12	Required, No crack & crazing
Chemical Resistance	ISO 10545.13	Required
Colour resistance to light	DIN 51094	No alteration
Staining Resistance	ISO 10545.14	Min. Class 3
Slippage Resistance	DIN 51130 DIN 51097	$\geq \text{R9}$ A

2. Approved Supplier or equivalent approved by the Architect

Arnhold & Co., Ltd.
 6/F Victoria Centre, 15 Watson Road, Hong Kong
 Tel: (852) 2806 4521 / 90905009
 Fax: (852) 2512 9986
 Email: raymondchan@arnhold.com.hk

E. Heavy Duty Porcelain Granite Tile

Origin/Brand : "KATANA" or approved equivalent
Texture : Anti-slip min. R11
Color : Light Grey GT62017 201221A
Size : 300x600
Thickness : 20 mm

1. Technical features : Complied with GB/T4100-2015 Annex G
 - a. Measurement of Length of tiles for $\pm 0.6\%$ of deviation from working size and $\pm 0.5\%$ of deviation from avg. size and complied with GB/T3810.2-2016
 - b. Determination of thickness of tiles and $\pm 5\%$ of deviation from working size and complied with GB/T3810.2-2016.
 - c. Determination of straightness of sides of tiles and $\pm 0.5\%$ of deviation from straightness and complied with GB/T3810.2-2016.
 - d. Determination of surfaced flatness of tiles and $\pm 0.5\%$ of deviation from working size and complied with GB/T3810.2-2016.
 - e. Determination of water absorption of tiles $E < 0.5\%$ and complied GB/T3810.3-2016.
 - f. Modulus of rupture avg. 45 (MPa) and complies with GB/T3810.4-2016
 - g. $S = 12061$ Breaking strength(N) and complies with GB/T3810.4-2016
 - h. Determination of stain resistance of tile min. level 3 and complied with GB/T3810.14-2016
 - i. Determination of abrasion resistance level 4 and complied with GB/T3810.7-2016
 - j. Class GB Resistance to acids and alkalis and complies with GB/T3810.13-2016
 - k. Determination of the coefficient of friction for floor tile should > 0.50 (Dry or Wet) and complied with GB/T4100-2015
 - l. Determination of the anti-slip property of floor tile should min. R11 and complied with DIN51130:2014.
2. Approved Supplier or equivalent approved by the Architect
K.A. Creative Building Materials Co., Ltd.
T: (852) 3589 6611 F: (852) 2117 1139
Contact Person: Ms. Peggy Fan
T: 9632 5647 | E-mail: info@kac.com.hk

F. Artificial Granite Tile

Origin/ Brand : RYOWA or approved equivalent
Texture : Texture "A" R11
Color : FT18A-KA43
Size : 592 x 292 x 18 mm (Recommend joint width 8mm *4 sides)
Thickness : 18 mm

1. Technical features : Complied with EN 14411:2006 (E) Annex G
 - a. Measurement of Length of tiles for $\pm 1.0\%$ of deviation from working size and $\pm 0.5\%$ of deviation from avg. size and complied with BS EN ISO 10545-2:1997

- b. Determination of thickness of tiles and $\pm 10\%$ of deviation from working size and complied with BS EN ISO 10545-2:1997
 - c. Determination of straightness of sides of tiles and $\pm 0.5\%$ of deviation from straightness and complied with BS EN ISO 10545-2:1997
 - d. Determination of surfaced flatness of tiles and $\pm 0.5\%$ of deviation from working size and complied with BS EN ISO 10545-2:1997
 - e. Determination of surface quality of tiles and minimum 95% of tiles without defects and complied with BS EN ISO 10545-2:1997
 - f. Determination of water absorption of tiles 0.5% and complied BS EN ISO 10545-3:1997
 - g. Determination of resistance to deep abrasion of tiles and max. 175 mm³ volume of material abraded and complied with BS EN ISO 10545-6:1997
 - h. Resistance to acids and alkalis and complied with BS EN ISO 10545-13:1997
 - i. R11 of texture "A" and slip resistance complies with DIN 51130
2. Approved Supplier or equivalent approved by the Architect

K.A. Creative Building Materials Co., Ltd.
T: (852) 3589 6611 F: (852) 2117 1139
Contact Person: Ms. Peggy Fan
T: 9632 5647 | E-mail: info@kac.com.hk

2.03 ANCILLARY MATERIALS

- A. "PCI" Tilefast and Waterproofing Systems or equivalent approved by the Architect, including Plastering, Waterproofing, Tile Adhesive, Grouting and Sealant:

Locations	Finishes	System Build Up
Flat Roof and Podium Deck	Tiles coverings	<u>Bonding slurry and spatterdash coats</u> "PCI Latex" (formerly known as "PCI Emulsion") modified synthetic resin dispersion base bonding slurry coat and spatterdash coats <u>Cement Sand Render and Screed</u> "PCI Latex" (formerly known as "PCI Emulsion") modified synthetic resin dispersion base C/S bedding mortar <u>Waterproofing coating</u> "DP Polyurea 3000" spray-applied fast cure Polyurea waterproofing membrane system (min. 1.5mm thick) (Refer to Section 07 10 00 Damproofing and waterproofing) <u>Tile Adhesive</u> "PCI Nanolight" deformable and flexible thin and medium-bed tile adhesive incorporating Nano technology <u>Tile Grout</u> "PCI Nanofug" universal flexible tile grout for every joint width incorporating Nano Technology <u>Joint Sealant</u> "PCI Silcofug E" elastic silicone rubber sealant containing fungicide and matching colour of specified joint grout (Refer to

		Section 07 92 00 Joint Sealants)
Water Tank	Tiles coverings	<u>Waterproof Concrete</u> “SCHOMBURG Betocrete C17 (BV)” Liquid Form Crystalline Waterproof Concrete Plasticizer including drinking water in accordance with DVGW worksheet W270 and W347 (Refer to Section 04 20 00 Unit Masonry) <u>Bonding slurry and spatterdash coats</u> “BASF Febond SBR” modified cement bonding slurry and spatterdash coats <u>Cement Sand Render and Screed</u> “BASF Febond SBR” modified C/S bedding mortar <u>Waterproofing coating</u> “DP Brushcrete” flexible cementitious waterproofing coating (min. 2mm thick) (Refer to Section 07 10 00 Dampproofing and waterproofing) <u>Tile Adhesive</u> “PCI FT-Klebemortel” non-toxic tile adhesive <u>Tile Grout</u> “PCI Durafug NT” non-toxic tile grout <u>Joint Sealant</u> “PCI Silcoferm KTW” silicone joint sealant in drinking water applications or equivalent complying with KTW and DVGW worksheet W270 requirements. (Refer to Section 07 92 00 Joint Sealants)
All internal wet area e.g. Lavatory / Plant room / Pump room / Refuse room and external wall with rendering finishes, etc.	Tiles coverings	<u>Bonding slurry and spatterdash coats</u> “PCI Latex” (formerly known as “PCI Emulsion”) modified synthetic resin dispersion base bonding slurry coat and spatterdash coats <u>Cement Sand Render and Screed</u> “PCI Latex” (formerly known as “PCI Emulsion”) modified synthetic resin dispersion base C/S bedding mortar <u>Waterproofing coating</u> “DP Brushcrete” flexible cementitious waterproofing coating (min. 1mm thick) (Refer to Section 07 10 00 Dampproofing and waterproofing) <u>Tile Adhesive</u> “PCI Nanolight” deformable and flexible thin and medium-bed tile adhesive incorporating Nano technology <u>Tile Grout</u> “PCI Nanofug” universal flexible tile grout for every joint width

		incorporating Nano Technology <u>Joint Sealant</u> “PCI Silcofug E” elastic silicone rubber sealant containing fungicide and matching colour of specified joint grout (Refer to Section 07 92 00 Joint Sealants)
For external wall with tiling system	Tiles coverings	<u>Bonding slurry and spatterdash coats</u> “PCI Latex” (formerly known as “PCI Emulsion”) modified synthetic resin dispersion base bonding slurry coat and spatterdash coats <u>Cement Sand Render and Screed</u> “PCI Latex” (formerly known as “PCI Emulsion”) modified synthetic resin dispersion base C/S bedding mortar <u>Waterproofing coating</u> “PCI Seccoral 2K Rapid” trowel-liquid applied flexible and crack-bridging anti-efflorescence barrier with waterproofing function (min. 1.5mm thick) (Refer to Section 07 10 00 Damproofing and waterproofing) <u>Tile Adhesive</u> “PCI Nanolight” deformable and flexible thin and medium-bed tile adhesive incorporating Nano technology <u>Tile Grout</u> “PCI Nanofug” universal flexible tile grout for every joint width incorporating Nano Technology <u>Joint Sealant</u> “PCI Silcofug E” elastic silicone rubber sealant containing fungicide and matching colour of specified joint grout (Refer to Section 07 92 00 Joint Sealants)

1. General Requirements of “PCI” tilefast and waterproofing systems including plastering, adhesive, grouting and sealant:
 - a. The proposed system must be deformable to compensate variation in temperature and elongation of substrates;
 - b. All components of the system are non-toxic and do not cause any hazard or injury to operatives or to others in the vicinity of the Site;
 - c. The system counting from the waterproofing membrane to the grouting shall prevent the cycle of efflorescence that normally occurs from the bedding mortar and substrate.
 - d. The Tile Adhesive shall be compatible with the Waterproofing Systems (refer to Section 071000) and shall not adversely affect the water-tightness performance of the latter. The compatibility shall be verified and tested by Construction Products list A, Part 2, no. 1.10 by DIBt. In no case shall the Tile Adhesive and tiling works invalidate, limit or alter the warranty of the Waterproofing Systems in whole or in part. Written confirmation from the specialist sub-contractor of the Waterproofing Systems in connection with

this requirement is at pre-requisite condition of the Architect's approval of any material submissions of the Tile Adhesive and Waterproofing System;

2. Manufacturer requirements

- a. The whole system must be from the same manufacturer/ supplier. All products must be compatible with each other within the same system;
 - 1) The manufacturer must have a minimum of 30 years experience in the tile-fixing business, specializing in the manufacture of high-performance, commercial grade Plastering, Tile Adhesive, Tile Grout and Joint Sealant System and materials.
 - 2) The manufacturer must have the quality standards of DIN EN ISO 9001, DIN EN ISO 14001 and OHSAS 18001
- B. Performance of "PCI Latex" (formerly known as "PCI Emulsion") modified synthetic resin dispersion base bonding slurry coat/spatterdash coats and bedding mortar :
 - a. Chemical nature - Aqueous dispersion of a polymer based on: acrylic polymers;
 - b. Vapour pressure: 23 hPa;
 - c. Information on water vapour pressure: 23 mbar (20°C);
 - d. pH value: 8.3 – 9.2 (20°C);
 - e. Possible Hazards (according to Directive 67/548/EWG or 1999/45/EC): No specific dangers known, if the regulations / notes for storage and handling are considered;
 - f. Can be used to modify mortar and render;
 - g. Plasticising, better workability and easier application of mortars and renders;
 - h. Help increase the wear resistance of mortar, less abrasion, thus longer solutions;
 - i. Improve bending tensile strength; help achieve a stress-free setting process, even with large areas;
 - j. Crack-free hardening, reduces movement in the mortar;
 - k. Minimum Flexural strength of emulsion modified render according to EN 196 Part 1 : 7.3 N/mm²;
 - l. Minimum Compressive strength of emulsion modified render according to EN 196 Part 1 : 23 N/mm² but less than 30 Mpa;
 - m. Capillary water absorption coefficient of emulsion modified render according to EN 13057: 0.70 kg / (m².h0.5);

C. Performance of “BASF Febond SBR” modified cement bonding slurry and spatterdash coats and bedding mortar:

				Normal Dose Range			
Quantity of latex in litres per 50kg cement used				10	12.5	15	20
Adhesion to concrete: Test Method 1	1W27D: DRY	N/mm ²		2.0		2.1	3.4
	1W20D7W: WET	N/mm ²		1.0		0.9	1.4
	1W20D63W: WET	N/mm ²		1.4		1.8	2.1
Adhesion to steel: Test Method 1	DRY	N/mm ²		2.0			1.6
	WET	N/mm ²					1.3
Slant Shear Adhesion (BS 6319 : Part 4) (Mortar / Febond SBR mortar using Febond SBR bonding slurry).							
Test Method 2:		N/mm ²		31			
Tensile strength (BS12):	DRY	N/mm ²					4.3
	WET	N/mm ²					3.9
Compressive strength (BS 1881:Part 4) :				0			
Flexural strength (BS 4551):	DRY	N/mm ²			6.2		10.6
	WET	N/mm ²					9.6
Effect of chemicals on Flexural Strength after 6 month immersion:							
Untreated		DRY N/mm ²			6.2		
10% Potassium Hydroxide		N/mm ²			12.3		
10% Magnesium Sulphate		N/mm ²			6.3		
5% Lactic Acid		N/mm ²			8.0		
10% Sucrose		N/mm ²			9.2		
5% Hydrochloric Acid		N/mm ²			2.2		
20% Ammonium Nitrate		N/mm ²			4.8		
Petroleum Spirit		N/mm ²			7.5		
Effect of Extremes of Temperature:							
Flexural strength:							
Untreated		N/mm ²					10.6
After 60 freeze-thaw cycles in 10% brine at -18°C		N/mm ²					10.4
After 1 year at 70°C		N/mm ²					14.3
Adhesion to Concrete:							
Untreated		N/mm ²					3.4
After 6 months at 70°C		N/mm ²					2.6
Thermal Expansion (Linear):							
from -20°C to +20°C		(Coeff x 106)				12.8	
from +20°C to +60°C		(Coeff x 106)				12.9	
Shrinkage during drying:							
Water / cement ratio				0.34		0.34	0.30
% shrinkage				0.02		0.01	0.01
Water Vapour Permeability (BS 4016 / BS 3177 11mm thick):				38.1		3.9	1.9
Water Penetration (30 metres per head of water was applied to one face of a mortar 15mm thick:				35		0	0
(i) with Febond SBR in mortar		kg / m ² / 24 hrs					
(ii) using Febond SBR / cement sealing coats on unmodified mortars							

Other Performance:	
Description	Values
pH	10.5
Specific gravity	1.01
Freeze-thaw stability	Withstands at least 5 freeze thaw cycles, but insides storage above 0°C recommended.
Particle size	0.17 micron*
Stabilization	Non-ionic
Butadiene content	45% by weight of the polymer
Antioxidant	Yes
Bacteriocide	Yes
Antifoam	Yes
Minimum film forming temperature	About 1°C
Shelf life (interior storage from above 0°C to 25°C	About 2 years, but agitate before use after prolonged storage
(* A cubic mm of dried film is made up from more than 125,000,000,000 particles	

- D. Performance of “PCI Nanolight” deformable and flexible thin and medium-bed tile adhesive incorporating Nano technology:
- a. Shall be pre-bagged with unique light filler and binding agent incorporating the Nano-technology to increase the contact surface down to nano level (10-9m) and to minimize the cavities inside the cement matrix and thus to develop better adhesion;
 - b. Be compatible with PCI Seccoral 2K / PCI Lastogum liquid applied waterproofing systems and have tested in accordance with Construction Products List A, Part 2, 1.10 by DIBt by Technical University Munich to verify their compatibility. The minimum requirements are the tensile adhesion strength between waterproofing and adhesive is greater than 0.7N/mm² even if they are stored under normal climate for 28 days, heat aging 70°C for 14 days, water immersion for at least 21 days, lime water Ca(OH)₂ or KOH for at least 7 days;
 - c. Be resistant to 1.5 bar (15 m height) water pressure when combined with PCI Seccoral / PCI Lastogum waterproofing system according to DIN 1048 Part 5;
 - d. Crack bridging to be 0.4 mm minimum when combined with PCI Seccoral / PCI Lastogum waterproofing system according to DIN 28052-6;
 - e. Low in chromates in compliance with TRGS 613;
 - f. Can be applied on all substrates: cementitious screeds, ground anhydrite and / or gypsum based floated screeds, concrete, precast concrete elements, aerated concrete, heated screeds, plaster boards, gypsum fibre boards, dry screeds, magnesite screeds, mastic asphalt (only indoors), insulation boards, plaster slabs, plaster, render and gauged render, masonry, old ceramic coverings and even, sound wooden substrates such as wooden chipboards, OSB boards etc., metal substrates (only indoors), firmly adhering PVC coverings.
 - g. Can be used to fix all ceramic coverings laid in thin or medium bed: non-vitrified tiles, vitrified tiles, fully vitrified tiles, porcelain mosaic, glass mosaic, glass tiles, brick tiles and terracotta.
 - h. High yield (30 – 40% better coverage than standard flexible adhesive mortar with the same weight) due to use of special additives and a unique filler combination (Approx. 800 kg/m³, i.e. 0.8kg of dry powder per m² and mm thickness of adhesive bed);
 - i. Classify as ‘EC1’ Plus R (90% dust-reduced and very low emission) according to GEV Emissioncode;
 - j. Rapid curing (tensile adhesion strength $\geq$ 1.0 N/mm² after 24 hours and walkable/groutable time after 5 hours), but still approx. 90 minutes working time and long open time (> 30 minutes);
 - k. Use PCI Nanolight adhesive for fixing various types of tiles by thin-bed and medium bed method up to a thickness of 10mm and with minimum 1mm;
 - l. Be used to level and repair the voids on surface and to fix the tiles directly on uneven substrate up to 10 mm thickness, for PCI Nanolight adhesive, in one time;
 - m. Meet the requirements of “Flexmortel” regulation issued by the German Building Construction Chemicals;
 - n. Be tested in accordance with the relevant type approval procedures of the Marine Equipment Director 96/98/EC and was found in compliance with the requirements of the international standards – IMP Res.A. 653(16); IMO Res. MSC.61(67), Annex I, Part 5;
 - o. Be tested in accordance with ISO 13007-1:2004-12 and been classified as ‘C2FTE S1’ for PCI Nanolight adhesive (C: Cementitious adhesive, 2: Improved performance of tensile adhesion strength, T: Reduced slip, E: Extended open time, F: Fast Setting, S1: Deformable adhesive); and having “CE” marking on each bag to declare the classification.
 - p. Be tested in accordance with EN 13501-1 : 2007 and has been classified as ‘A2-S1’ and ‘d0’;

- q. Classify as 'A2' according to DIN4102:
- r. Be tested in accordance with EN12002 and classified as 'S2' (highly deformable adhesive) when mix with PCI Lastoflex and therefore having the following performance:
 - (a) Tensile Adhesion Strength after 28h r.t. (EN 1348) $\geq 2.5 \text{ N/mm}^2$
 - (b) Tensile Adhesion Strength after water immersion (EN 1348) $\geq 1.7 \text{ N/mm}^2$
 - (c) Tensile Adhesion Strength after heat ageing (EN 1348) $\geq 3.6 \text{ N/mm}^2$
 - (d) Tensile Adhesion Strength after freeze-thaw cycles. (EN 1348) $\geq 1.5 \text{ N/mm}^2$
 - (e) Extended Open Time after 30 mins (EN 1346) $\geq 1.2 \text{ N/mm}^2$
- s. Resistant to efflorescence when it is combined with PCI Emulsion and PCI Dichtungsmittel additives, PCI Seccoral 2K Waterproofing and PCI tile grout;
- t. Tile Adhesive must also demonstrate its compatibility according to ZDB leaflet and DIBt "Instructions for the installation of waterproofing directly under tile coverings for interior and exterior use", in toilets, showers and swimming pool areas.
- u. Third Party Verified EPD according to ISO 14025 and EN15804.

E. Performance of “PCI FT-Klebemortel” non-toxic tile adhesive:

- a. Shall be factory pre-bagged in dry powder form. On site mixing with liquid component from elastifying polymer/latex and filler is strongly prohibited;
- b. Be compatible with PCI Lastogum liquid applied waterproofing system and have tested in accordance with PG-AIV (a general building inspection test certificate for liquid-applied waterproofing material for use with tiles and slab) by Technical University Munich to verify their compatibility. The minimum requirements of tensile adhesion strength between the waterproofing and the tile adhesive is a) ≥ 0.8 N/mm² under normal climate, b) ≥ 1 N/mm² under water immersion, c) ≥ 0.9 N/mm² under heat aging, d) ≥ 1.3 N/mm² under lime water Ca(OH)₂;
- c. Be resistant to 1.5 bar (15m height) water pressure when combined with PCI Lastogum waterproofing system according to DIN 1048 Part 5;;
- d. Can be applied on cement screeds, rough-ground anhydrite and/or gypsum based floated screeds, renders, concrete, heated screeds, plaster boards, gypsum fibre boards, plaster slabs and plaster;
- e. Can be laid in thin bed method: non-vitrified tiles, vitrified tiles and fully vitrified tiles;
- f. Can lay ceramic tiles in areas exposed to permanent wetness as well as in wet areas such as showers, toilets, bathrooms, and cold storage rooms;
- g. Be non-combustible and classified as Grade A1 in accordance with EN13501-1:2007;
- h. Good adhesion without pre-wetting or priming on concrete, sand-cement screed and render;
- i. Be tested in accordance with EN12004:2007 and classified as ‘C2T’ (C: Cementitious adhesive, 2: Improved performance of tensile adhesion strength, T: Reduced slip) and having “CE” marking on each bag to declare the classification;
- j. Minimum adhesion after dry storage is 1.53 N/mm² according to DIN EN 1348:2007-11;
- k. Minimum adhesion after water immersion is 1.19 N/mm² according to DIN EN 1348:2007-11;
- l. Minimum adhesion after heat aging is 1.29 N/mm² according to DIN EN 1348:2007-11;
- m. Minimum adhesion after freeze-thaw storage is 1.39 N/mm² according to DIN EN 1348:2007-11;
- n. Minimum 20 min. Open time: tensile adhesion strength is 1.12 N/mm² according to DIN EN 1348:2007-11;
- o. Minimum 30 min. Open time: tensile adhesion strength is 0.85 N/mm² according to DIN EN 1348:2007-11;
- p. Maximum vertical slippage is 0.20mm according to DIN EN 1348:2007-11;
- q. Shall not contain asbestos or other mineral fibres;
- r. Shall not omit injurious silica dust during the application;
- s. Conform to the requirements for the use in the area of drinking water pursuant to DVGW work sheet W 270 and DVGW work sheet W 347.

F. Performance of “PCI Nanofug” universal flexible tile grout for every joint width incorporating Nano Technology:

- a. Shall be pre-bagged with unique binding agent incorporating the nano-technology to increase the contact surface down to nano level (10-9m) and to minimize the cavities inside the cement matrix and thus to develop very low adsorption;
- b. Surface repellent to water and dirt;
- c. Easy to clean effect, due to special additives;
- d. Be non-combustible and classified as Grade A2 according to DIN 4102;
- e. Low in chromates in compliance with TRGS 613;
- f. Conform to DVGW sheet W270;
- g. Confirm to Marine Equipment Directive 96/98/EC and to type as described in the EC Type Examination Certificate, especially when combined with PCI Seccoral 1K / PCI Lastogum waterproofing and PCI Nanolight tile adhesive;
- h. Accommodate large range of joint width from 1 to 10mm and from 0mm upwards can be blended with 25% of 0.1-0.4mm silica sand;
- i. Rainproof after approx. 4 hours at 23°C and 50% R.H;
- j. Resistant to mould growth in accordance with BS 5980, Appendix B;
- k. Be tested in accordance with EN 13888: 2009 and classified as ‘CG2 WA’;
- l. Be tested in accordance with EN 12808, and the flexural strength and compressive must not be less than 6 N/mm² and 24 N/mm², respectively;
- m. Be tested in accordance with ANSI A118.6 Clause H4.3 to have the linear shrinkage of less than 0.1% and 0.2% after 1 day and 7 days, respectively, under 23°C and 50% R.H., and be tested in accordance with EN 12808 Part 4 to have the linear shrinkage of less than 1.5mm/m after 28 days;
- n. Be tested in accordance with ANSI A118.6 Clause H3.4 to have the water absorption rate of less than 5% (50% R.H. to immersion) and 7% (immersion to dry), under 23°C and 50% R.H., and be tested in accordance with EN 12808 Part 5 to have the water absorption < 1g and < 2g after 30min. and 240 min., respectively;
- o. No need to on site mixing with liquid from elastifying polymer or latex as it is factory pre-bagged in dry powder form.
- p. Available Colours are no. 02 Bahamabeige, no. 03 Caramel, no. 05 Mittelbraun, no. 11 Jasmin, no. 12 Anemone, no. 16 Silbergrau, no. 41 Dunkelbraun, no. 18 Manhattan, no. 19 Basalt, no. 20 Weiß, no. 21 Hellgrau, no. 23 Lichtgrau, no. 31 Zementgrau, no. 40 Schwarz, no. 43 Pergamon, no. 44 Topas and no. 47 Anthrazit; Colour to be approved by Architect.

G. Performance of “PCI DuraFug NT” non-toxic tile grout

- a. Shall be latex pre-bagged colour grout with unique binding agent combination incorporating the Nano technology to increase the contact surface down to nano level (10-9m) and to minimize the cavities inside the cement matrix and thus to develop very low adsorption for all every joint widths;
- b. High resistance to mechanical stress by nanofill technology;
- c. For joint widths from 3 to 20 mm;
- d. High-density texture of grout guarantees minimal depth of penetration of pollution and high mechanical load;
- e. Resistant to de-icing salt;
- f. Increased resistance to acid cleaners compared to usual cementitious joint grouts;
- g. Resistant to stress caused by high-pressure water jet cleaning up to 100 bar;
- h. Crack-free hardening, the joint does not crack or crumble;
- i. Waterproof and frost resistant, multi-purpose use indoors and out;
- j. With official test certificate:
 - for potable water application according to the DVGW worksheets W347 and W270.
 - official test certificate – fire class A1 (DIN 4102)
- k. Temperature resistance is -20°C to +250°C;
- l. Resistance to Chemicals

Resistance		Resistance	
Organic acids		Solvents	
Formic acid	(+)	Acetone	+
Acetic acid	(+)	Butyl alcohol	+
Oxalic acid	+	Ethanol	+
Tartaric acid	+	Isopropanol	+
Citric acid	(+)	Petroleum ether	+
		Formaldehyde	+
		Xylene	+
Lyes		Others	
Ammonia	+	Petrol	+
Calcium hydrate solution	+	Developer solution	+
Chlorine bleaching	+	Glycerine	+
Caustic potash solution	+	Whey	(+)
Caustic soda solution	+	Hydrogen peroxide	+
Oils			
Fuel oil	+		
Paraffin oil	+		
Hydraulic oil	+		
Silicone oil	+		
Edible oil	+		
Turpentine oil	+		

Legend:

+ = resistant, (+) = short-term resistant

Test Methods	Test result
EN 12808 Part 2 Resistance to abrasion (mm ³)	≤150
EN 12808 Part 3 Flexural strength (N/mm ²) Compressive strength (N/mm ²) (after 28d under Standard Conditions)	≥9 ≥68
EN 12808 Part 3 Flexural strength (N/mm ²) Compressive strength (N/mm ²) (after Freeze-Thaw-Immersion)	≥4 ≥90
EN 12808 Part 4 Shrinkage (mm/m)	≤1.4
EN 12808 Part 5 Water absorption after 30 Minutes (g) Water absorption after 240 Minutes (g)	≤0.6 ≤1.2

- m. Classify as improved resistance to abrasion and water absorption (CG2) tile grout in accordance with EN 13888 (12.2002);
- n. Be tested in according with the relevant type approval procedures of the Marine Equipment Director 96/98/EC and was found in compliance with the requirements of the international standards – IMP Res.A. 653(16); IMO Res. MSC.61(67), Annex I, Part 5.

PART 3 EXECUTION

3.01 EXAMINATION/INSPECTION

- A. Verify that surfaces to be tiled fall within the maximum variations shown below:

	Walls	Floors
Dry-Set Mortar:	1.3mm in 1 metre	1.0mm in 1 metre
Epoxy:	1.3mm in 1 metre	1.0mm in 1 metre
Organic Adhesive:	1.3mm in 1 metre	1.7mm in 1 metre

- B. Report all unacceptable surfaces to the Architect and do not tile such surfaces until they are levelled enough to meet above requirements. Leveling coat shall be included in this Section.

3.02 PREPARATION

- A. Protect surrounding work from damage or disfiguration.
- B. Vacuum clean existing substrate and damp clean.
- C. Ensure surfaces to be tiled are free of curing membranes, oil, grease, wax and dust.
- D. Seal substrate surface cracks with filler. Level existing substrate surfaces to acceptable flatness tolerances.
- E. Prepare substrate surfaces with sealers or conditioners as recommended by the Manufacturer.
- F. Surface Preparation
1. Check if the surface to be tiled is clean (no contamination, such as oil or grease) and flat, with no cracks larger than 6 mm. Correct any deficiencies. It may be necessary to first install an “anti-fracture” membrane.
 2. Do not set over new concrete that is less than 28 days from pour. Setting tile over other materials, e.g., wood substrate or install cement board, seek manufacturer’s installation instructions.

3.03 INSTALLATION/APPLICATION

- A. Layout:
1. Determine locations of all movement joints before starting tilework.
 2. Lay out all tilework as shown on the Drawings with no cut tiles. Cut tiles are only permitted where specifically noted on the drawings.
 3. Lay out tile wainscots to next full tile beyond dimension shown.
 4. Align all wall joints to give straight uniform grout lines, plumb and level.
 5. Align all floor joints to give straight uniform grout lines.
- B. Expansion Joint Installation:
1. Expansion joints in tile and paver shall be located over all cold joints and saw-cut control joints.

2. Joints in tile, and paver and setting materials shall never be less than the width of the saw-cut control joint width. Refer to Section 6.5 for sealant joint installation requirements.
3. To insure that location of joints in tile work align with existing joints in substrate, joints in tile work should be constructed during installation of mortar beds, paver and/or tile, rather than saw-cutting joints after installation.
4. Keep expansion joint cavities open and free of dirt, debris, grout, mortar and setting materials, incorporate sealant joint along the interface.
5. Set compressible back-up strip when mortar is placed or utilize removable wood strip to provide space for back-up after mortar has cured.
6. Install sealant after tile and paver work and grout are dry. Follow Sealant Manufacturer's recommendations.
7. Refer to sealant section in ANSI / BS tile installation specification, and Section 6.5
8. Verify with manufacturer the recommendations for the maximum spacing of expansion joints to the tiles. For guidance only, the following shall be used:
 - 1) Interior tilework - 7.3 to 11 metres in each direction.
 - 2) Exterior tilework - 3.6 to 4.9 metres in each direction.
 - 3) Interior tilework (exposed to direct sunlight or moisture) - 3.6 to 4.9 metres in each direction.
9. Where tilework abuts restraining surfaces such as perimeter walls, dissimilar floors, curbs, columns, pipes, ceilings, and where changes occur in backing materials, incorporate sealant joint along the interface.
10. Tile edges to which sealant will bond to must be clean and dry, sand or grind as necessary to obtain optimum sealant bond.
11. Prime tile edges when recommended by the Sealant Manufacturer. Take care to keep primer off tile faces. Install sealant after tilework and grout are dry. Follow Manufacturer's recommendations. All expansion joint sealants shall be tooled.

C. Tile Installation:

1. The setting bed material is to be chosen from materials specified above only. The substrate and site conditions are to be reviewed with the tile and adhesive manufacturer and their recommendations transmitted to the Architect for final review.

2. Installation Methods:-

- a. In accordance with Manufacturer's recommendations and BS 5385. Render / screed shall be completed and cured at least 2 weeks before tiling (for swimming pool / water tank case at least 3 weeks).
- b. Lay tile to pattern indicated. Verify that pattern is uninterrupted through openings.
- c. Fit tile tight to protrusions and vertical interruptions. Form corners and bases neatly.
- d. Work tile joints uniform in width, subject to variance in tolerance allowed in tile size. Joints shall be watertight, without voids, cracks, excess mortar, or grout.
- e. Sound tile after setting. Replace hollow sounding units.
- f. Keep expansion/contraction control joints free of mortar or grout.
- g. Allow tile to set for a minimum of 48 hours prior to grouting.
- h. Install grout from the same batch in one complete area. Do not mix grout batches.
- i. Two pack resin based organic materials (notably epoxy resin) should be used as tile grouting for internal wet areas and especially for floors that must be resistant to acids, or when the floor is to be smoothed successively, since the resulting inter-tile spaces will be compact and without cracks; whenever resins are used for grouting always make sure that excess adhesive is cleaned off the tile surface before it hardens. The contractor shall submit a suitable proprietary colour grouting for the architect's approval.
- j. For wall tile, cementitious base adhesive shall be utilized for load bearing structures in well set concrete; when dealing, however, with elastic structures or structures that are sensitive to shrinkage or expansion caused by humidity fairly elastic adhesives shall be used. In the latter case it is important to provide numerous and efficient expansion and shrinkage joints in order to avoid, as far as possible, damage resulting from the afore-mentioned tile movements. For correct use of adhesives, the contractor shall follow manufacturers' instructions scrupulously. The contractor shall submit a suitable proprietary adhesive for the architect's approval.

3.04 CLEANING

- A. All tilework shall be clean of grout film upon completion. Use of abrasive cleaners is strictly prohibited.

3.05 PROTECTION

- A. Cover all tile floors with a heavy-duty, non-staining construction paper, taped in place.
- B. Prohibit all foot and wheel traffic from using newly tiled floors for at least 3 - 7 days.
- C. Place large, flat cushioned boards in walkways and wheelways after 7 days where use of newly tiled floor with cement type grout is unavoidable.
- D. After completion of tile work to all toilet areas, these rooms shall be kept locked and entry prohibited until Project Handover.

E. Surface Levels of Payment Courses

1. The surface level of the pavement at any point shall not deviate vertically from the true finished surface by more than $\pm 6\text{mm}$.
2. Pavement shall be measured transversely for irregularities at points decided by the Architect, by a 3m long straight-edge. The maximum deviation of the pavement surface below the straight-edge shall not exceed 3mm.

F. Rectification Work

Where any tolerances in this specification are exceeded the Contractor shall in conjunction with the Architect determine the full extent of the area which is out of tolerance and shall make good at his expense the surface of the pavement course or formation in the manner agreed entirely to the satisfaction of the Architect.

END OF SECTION